AWS Cloud Practitioner Exam: Cheat Sheet, AWS Exam Prep Tips & Elimination Strategies

Aws Services Cheat Sheet

Compute Services

- **EC2 (Elastic Compute Cloud)**: Virtual servers in the cloud, offering scalable compute capacity.
- **EC2 Auto Scaling**: Automatically adjusts the number of EC2 instances based on traffic.
- **Lambda**: Serverless compute service that runs code in response to events without provisioning servers.
- **Elastic Beanstalk**: PaaS for deploying and managing applications in various programming languages.
- **Lightsail**: Simplified virtual servers for simpler web applications.
- **Elastic Container Service (ECS)**: Managed service for running Docker containers.
- **Elastic Kubernetes Service (EKS)**: Managed Kubernetes service for container orchestration.
- **AWS Fargate**: Serverless compute engine for containers.
- **AWS Batch**: Managed batch processing service for running large-scale parallel and high-performance computing workloads.

Storage Services

- **S3 (Simple Storage Service)**: Object storage with high durability and scalability.

- **EBS (Elastic Block Store)**: Block storage for EC2 instances.
- **EFS (Elastic File System)**: Scalable file storage for Linux-based workloads.
- **Glacier**: Low-cost archival storage for long-term data.
- **S3 Glacier Deep Archive**: Extremely low-cost archival storage with retrieval times in hours.
- **Storage Gateway**: Hybrid cloud storage service that integrates on-premises environments with cloud storage.
- **FSx for Windows File Server**: Managed Windows file system.
- **FSx for Lustre**: Managed file system for compute-intensive workloads.
- **S3 Transfer Acceleration**: Speeds up file uploads to S3.

Database Services

- **RDS (Relational Database Service)**: Managed relational databases (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server).
- **Aurora**: High-performance managed relational database compatible with MySQL and PostgreSQL.
- **DynamoDB**: Fully managed NoSQL database for key-value and document data.
- **Redshift**: Data warehousing service for big data analytics.
- **ElastiCache**: In-memory caching service (supports Redis and Memcached).
- **DocumentDB**: Managed NoSQL document database service, compatible with MongoDB.
- **Neptune**: Managed graph database service.
- **Timestream**: Managed time-series database for IoT and operational data.
- **Qldb (Quantum Ledger Database)**: Immutable ledger database.
- **Aurora Serverless**: On-demand, auto-scaling version of Aurora.

Networking Services

- **VPC (Virtual Private Cloud)**: Isolated cloud network for resources.
- **Route 53**: DNS and domain registration service.

- **CloudFront**: Content Delivery Network (CDN) for fast content delivery.
- **Elastic Load Balancer (ELB)**: Distributes incoming application traffic across multiple targets (e.g., EC2 instances).
- **Direct Connect**: Dedicated network connection between on-premises data centers and AWS.
- **Transit Gateway**: Centralized hub to connect VPCs and on-premises networks.
- **VPC Peering**: Connects two VPCs directly.
- **PrivateLink**: Securely access services over AWS's private network.
- **Global Accelerator**: Improves application availability and performance for global users.

Security & Identity Services

- **IAM (Identity and Access Management)**: Manage user access and permissions to AWS resources.
- **AWS Shield**: DDoS protection service.
- **AWS WAF (Web Application Firewall)**: Protects web applications from common web exploits.
- **KMS (Key Management Service)**: Manages encryption keys for securing data.
- **Cognito**: User authentication and identity management for apps.
- **GuardDuty**: Threat detection service that continuously monitors for malicious activity.
- **Inspector**: Security assessment service to find vulnerabilities in EC2 instances.
- **Macie**: Machine learning-powered data security service for discovering sensitive data in S3.
- **Secrets Manager**: Securely stores and manages sensitive information like database credentials.

Developer Tools

- **CodeCommit**: Git-based version control service.
- **CodePipeline**: Continuous integration and continuous delivery (CI/CD) service.
- **CodeBuild**: Build and test code in the cloud.
- **CodeDeploy**: Automates application deployment to EC2 instances and Lambda.
- **Cloud9**: Cloud-based IDE for writing, running, and debugging code.
- **X-Ray**: Helps analyze and debug distributed applications.
- **AWS SAM (Serverless Application Model)**: Framework for building serverless applications.

Management & Governance Services

- **CloudWatch**: Monitoring and observability service for AWS resources.
- **CloudTrail**: Tracks user activity and API usage across AWS infrastructure.
- **AWS Config**: Monitors and records AWS resource configurations.
- **AWS Organizations**: Manage multiple AWS accounts and apply policies across accounts.
- **AWS Control Tower**: Set up and govern a secure, multi-account AWS environment.
- **Service Catalog**: Create and manage catalogs of approved IT services for AWS resources.
- **AWS Systems Manager**: Unified interface for managing AWS resources and automating tasks.
- **AWS Trusted Advisor**: Provides recommendations to optimize your AWS environment.

Machine Learning Services

- **SageMaker**: Build, train, and deploy machine learning models.
- **Rekognition**: Image and video analysis using machine learning.

- **Comprehend**: Natural language processing (NLP) service for analyzing text.
- **Lex**: Build conversational interfaces (chatbots) using voice and text.
- **Polly**: Text-to-speech service.
- **Translate**: Language translation service.
- **Transcribe**: Speech-to-text service.
- **Forecast**: Time-series forecasting for demand planning and resource optimization.
- **Personalize**: Create personalized recommendations for users.

Analytics Services

- **Athena**: Query data in S3 using standard SQL.
- **EMR (Elastic MapReduce)**: Big data processing using Hadoop and Spark.
- **Glue**: Managed ETL (Extract, Transform, Load) service for preparing data for analytics.
- **Kinesis**: Real-time data streaming and analytics.
- **QuickSight**: Business intelligence service for data visualization.
- **Data Pipeline**: Orchestrates data workflows for processing and moving data.
- **Kinesis Data Firehose**: Real-time data delivery service to S3, Redshift, or Elasticsearch.
- **Kinesis Video Streams**: Real-time video stream processing.
- **Lake Formation**: Simplifies the process of setting up and managing data lakes.

Migration & Transfer Services

- **DMS (Database Migration Service)**: Migrate databases to AWS.
- **Server Migration Service (SMS)**: Migrate virtual machines to AWS.
- **Application Discovery Service**: Helps plan migration by discovering on-premises applications.
- **Snowball**: Physical device for transferring large amounts of data to AWS.

- **DataSync**: Transfer large data between on-premises storage and AWS.

Cost Management Services

- **AWS Budgets**: Set custom cost and usage budgets.
- **Cost Explorer**: Visualize and analyze AWS costs and usage.
- **Pricing Calculator**: Estimate AWS service costs based on usage.
- **Cost Anomaly Detection**: Detect unusual spending patterns.
- **AWS Cost and Usage Report**: Detailed CSV report of your AWS usage and costs.

Hybrid Cloud Services

- **VMware Cloud on AWS**: Run VMware workloads on AWS infrastructure.
- **Outposts**: Fully managed AWS infrastructure for on-premises environments.
- **AWS Snowcone**: Portable edge computing device for data collection and transfer.

Other Key Services

- **SQS (Simple Queue Service)**: Managed message queuing service.
- **SNS (Simple Notification Service)**: Publish/subscribe messaging service.
- **CloudFormation**: Infrastructure as code service for automating resource provisioning.
- **Elastic Transcoder**: A fully managed media transcoding service by AWS that converts video and audio files into formats compatible with various devices.
- **AWS Well-Architected Tool**: Helps review workloads against AWS best practices.

AWS Exam Prep Tips

To help you prepare for the AWS Certified Cloud Practitioner exam, here are some tips that can increase your chances of success:

1. Understand the Exam Objectives

The AWS Certified Cloud Practitioner exam tests your understanding of AWS Cloud concepts, services, pricing, and architecture. Key areas include:

- **Cloud Concepts**: Understand cloud computing, its benefits, and deployment models (Public, Private, Hybrid).
- **AWS Core Services**: Know the main AWS services for compute, storage, databases, networking, security, and machine learning.
- **AWS Pricing and Billing**: Learn about AWS pricing models, the Free Tier, and AWS Cost Management tools.
- **Cloud Security**: Understand security best practices, AWS shared responsibility model, IAM, and encryption.
- **AWS Global Infrastructure**: Learn about AWS regions, availability zones, edge locations, and the global network.

2. Use the AWS Free Tier for Hands-on Practice

One of the best ways to prepare is by getting hands-on experience with AWS services. The AWS Free Tier offers free access to a range of AWS services with usage limits. This will allow you to explore services like EC2, S3, Lambda, and more without incurring charges. Key services to explore:

- **EC2**: Launch and manage virtual servers.
- **S3**: Create and manage storage buckets.
- **IAM**: Set up users, groups, and roles to understand access control.

- **CloudWatch**: Set up basic monitoring for your AWS resources.
- **Lambda**: Write serverless functions and trigger them with events.

3. Focus on Key AWS Services

While the exam covers a broad range of services, it's essential to focus on the most commonly tested services:

- **Compute**: EC2, Lambda, Elastic Beanstalk
- **Storage**: S3, EBS, Glacier, EFS
- **Databases**: RDS, DynamoDB, Aurora
- **Networking**: VPC, Route 53, CloudFront, ELB
- **Security**: IAM, KMS, Shield, WAF
- **Cost Management**: AWS Pricing Calculator, AWS Budgets, Cost Explorer

4. Understand AWS Pricing and Billing

A significant portion of the exam covers AWS pricing and billing. Focus on the following:

- **AWS Pricing Models**: On-Demand, Reserved, and Spot Instances for EC2.
- **Free Tier**: What services are available in the Free Tier and their usage limits.
- **Cost Management Tools**: AWS Pricing Calculator, AWS Budgets, and AWS Cost Explorer.
- **Support Plans**: Understand the different support plans (Basic, Developer, Business, Enterprise) and their offerings.

5. Master the AWS Shared Responsibility Model

The AWS Shared Responsibility Model is a critical concept. It defines the division of security responsibilities between AWS and the customer:

- **AWS Responsibility**: Security of the cloud (physical security, infrastructure, etc.).
- **Customer Responsibility**: Security in the cloud (data encryption, access control, etc.).

6. Review the AWS Well-Architected Framework

The Well-Architected Framework is a set of best practices for building secure, high-performing, resilient, and efficient infrastructure for applications. It consists of 5 pillars:

- **Operational Excellence**: Focus on monitoring, automation, and improving processes.
- **Security**: Protect data and control access using encryption, identity management, and secure infrastructure.
- **Reliability**: Ensure your system can recover from failures and scale with strategies like multi-AZ and disaster recovery.
- **Performance Efficiency**: Use the right resources for your needs and scale as required.
- **Cost Optimization**: Manage costs by optimizing resource usage and selecting the right pricing options.

7. Take Practice Exams

Practice exams are an excellent way to gauge your knowledge and get a feel for the types of questions you will encounter on the actual exam. AWS offers a practice exam for the Cloud Practitioner certification, which mimics the format and difficulty of the real exam.

8. Manage Your Time During the Exam

The AWS Cloud Practitioner exam consists of 65 questions, and you have 90 minutes to complete it. Here are some time management tips:

- **Read each question carefully**: Understand the question and eliminate obvious wrong answers.
- **Pace yourself**: Don't spend too much time on any one question. If unsure, mark it and move on, then come back to it later.
- **Use the process of elimination**: Eliminate clearly incorrect options and make an educated guess.

9. Stay Updated with AWS Announcements

AWS regularly launches new services and updates existing ones. Stay updated with the latest announcements to ensure you are aware of any new features or services that may be covered in the exam.

10. Relax and Stay Confident

On the day of the exam, ensure you're well-rested and confident. Trust your preparation and knowledge. Don't rush through the exam, and take your time to review your answers before submitting.

Breakdown of the AWS Certified Cloud Practitioner Exam Domains

Domain	Weight	Number of Questions
Cloud Concepts	26%	17 questions
Security and Compliance	25%	16 questions
Technology	33%	21 questions
Billing and Pricing	16%	11 questions

Key Focus Areas for Each Domain:

- **Cloud Concepts**: Understand basic cloud terms and AWS value.
- **Security and Compliance**: Focus on security models, IAM, and compliance.
- **Technology**: Focus on AWS services, architecture best practices, and the AWS Well-Architected Framework.
- **Billing and Pricing**: Learn AWS pricing models and cost management tools.

Elimination Strategies

Rule Out Irrelevant Options

- Carefully analyze the question's context and eliminate services that don't fit.
  - *Example:* If the question is about analytics, exclude storage services like Amazon S3.

Avoid Distractors

- Watch out for answers that seem plausible but aren't entirely correct.
  - *Example:* For a serverless database, exclude Amazon RDS, which requires managing instances, and opt for Amazon Aurora Serverless.

Eliminate Outdated Services

- Choose modern AWS services over deprecated ones.
  - *Example:* For real-time data streaming, choose Amazon Kinesis over older tools like Elastic MapReduce (EMR).

Focus on AWS Best Practices

- Align answers with AWS Well-Architected principles like scalability, security, and cost-effectiveness.
    - *Example:* For cost optimization, avoid over-provisioned services like EC2 instances and choose Spot Instances.

Eliminate Non-AWS Solutions

- If the question is AWS-specific, exclude non-AWS tools.
    - *Example:* For monitoring, avoid third-party tools like Nagios unless specified.

Rule Out Manual Overhead Options

- AWS emphasizes automation and managed services. Discard solutions requiring manual effort.
    - *Example:* For infrastructure provisioning, prefer AWS CloudFormation over manual CLI commands.

Eliminate Single-AZ Options for High Availability

- For high availability, avoid single-AZ solutions.
    - *Example:* For databases, choose Multi-AZ RDS over single-instance deployments.

Exclude Overly Broad or Generic Answers

- Choose the most specific service for the task.
    - *Example:* For global load balancing, prefer Amazon Route 53 instead of a generic load balancer.

Watch for Misleading Terminology

- Ensure that AWS terminology is used correctly.
    - *Example:* "Amazon S3 for low-latency transactional databases" is incorrect because S3 isn’t designed for low-latency database use.

Key Points to Choose Correct Answers

Understand Service Categories

- Match the service to the category (compute, storage, networking, etc.).
  - *Example:* For a CDN, the correct answer is likely Amazon CloudFront.

Look for Keywords

- Identify keywords like "scalable," "managed," or "low-latency."
  - *Example:* For serverless compute, AWS Lambda is the answer.

Global vs. Regional Services

- Consider whether the service operates globally or regionally.
  - *Example:* For global delivery, choose CloudFront over S3.

Pay Attention to Pricing Models

- Understand AWS pricing models (pay-as-you-go, spot instances, etc.).
  - *Example:* For variable workloads, choose Spot Instances for cost efficiency.

Compliance and Regulatory Requirements

- If compliance is mentioned, pick services with certifications or compliance features.
  - *Example:* For GDPR, choose services offering data residency controls like Amazon S3.

Networking and Connectivity

- Evaluate networking-related options carefully.
  - *Example:* For secure private connections, choose AWS Direct Connect.

Workload-Specific Optimization

- Choose services tailored for specific workloads.

    - *Example:* For containerized applications, choose ECS or EKS over EC2.

Event-Driven Architectures

- For event-driven use cases, focus on services like SNS, SQS, or EventBridge.
    - *Example:* For decoupling microservices, choose SQS over direct service-to-service communication.

Data Analytics and Insights

- Choose services based on the type of analytics required.
    - *Example:* For big data processing, choose Amazon EMR; for real-time analytics, use Kinesis.

Security Enhancements

- Prioritize services with built-in security features like encryption and access control.
    - *Example:* For secure key management, choose AWS KMS.

DevOps and CI/CD Pipelines

- For CI/CD automation, choose services like CodePipeline, CodeBuild, and CodeDeploy.
    - *Example:* For fully automated CI/CD, avoid manual build steps.

Storage Access Patterns

- Choose storage based on access frequency and requirements.
    - *Example:* For frequently accessed data, choose S3 Standard; for infrequent access, use S3 IA.

Machine Learning and AI

- For ML workloads, choose services that streamline model training and deployment.

- *Example:* Use Amazon SageMaker for end-to-end machine learning workflows.

Backup and Disaster Recovery

- Choose services with automated backup and disaster recovery features.
  - *Example:* Use AWS Backup for centralized backup management.

Specialized Tips for AWS Questions

Architectural Design Questions

- Focus on AWS Well-Architected Framework, emphasizing scalability, security, and cost optimization.

Migration Scenarios

- For database migrations, choose AWS DMS or Migration Hub over manual methods.

Monitoring and Logging

- Prefer AWS-native solutions like CloudWatch and CloudTrail for monitoring and logging.

Disaster Recovery

- Select services with automated backups and multi-region replication, such as RDS with cross-region read replicas.

By following these elimination strategies and key considerations, you can effectively navigate AWS-related questions and identify the most appropriate services.

With a solid understanding of AWS services, hands-on experience, and practice exams, you'll be well on your way to passing the AWS Certified Cloud Practitioner exam. Good luck!